OOP, Models & DTOs

Encapsulation, domain models, and typed API responses

Agenda

1. **OOP & Encapsulation** – why `any` is dangerous, and how classes fix it
2. **Models (MVC)** – interfaces vs classes, domain models vs DTOs
3. **Putting it Together** – typed service, DTO response, updated controller

Part 1

OOP & Encapsulation

Where We Left Off

In Part 3, to keep things simple, the service used `any` for its data:

```ts
// src/services/postService.ts
const mockPosts: any[] = [
  { id: 1, title: "Getting Started", content: "TypeScript basics...", author: "Alice" },
];

export class PostService {
  async findById(id: number): Promise<any> {
    return mockPosts.find(p => p.id === id);
  }
}
```

This works – but it throws away all the benefits of TypeScript.

Why `any` is Bad Practice

```
// TypeScript has no idea what shape this data is
async getById(req: Request, res: Response): Promise<void> {
  const post: any = await this.service.findById(Number(req.params.id));

  console.log(post.tile);    // ❌ typo — no error, undefined at runtime
  console.log(post.Author);  // ❌ wrong case — silently undefined
  res.json(post);             // ❌ could expose any field, including sensitive ones
}
```

Using `any`:

- **Disables type checking** – TypeScript can't catch typos or wrong field names
- **Hides intent** – readers can't tell what shape the data has
- **No IDE help** – no autocomplete, no refactoring support

> `any` is an escape hatch – not a solution.

The Fix Starts with a Shape

The first step is giving your data a **known shape** using an interface:

```
// src/models/post.ts
export interface Post {
  id: number;
  title: string;
  content: string;
  author: string;
}
```

Now TypeScript can **catch typos** at compile time, **provide autocomplete**, and **document intent** – any reader knows exactly what a `Post` looks like.

But interfaces alone can still describe **invalid states** – nothing stops `id: -1` or `title: ""`. That's where classes come in.

Classes vs Interfaces

	Interface	Class
Purpose	Describe a shape	Implement behaviour
Can have methods	✅ (signatures only)	✅ (with implementation)
Can enforce rules	❌	✅
Compiled to JS	❌ (erased)	✅
Can be `new` -ed	❌	✅

> Use interfaces for **contracts**. Use classes when you need **behaviour and rules**.

What is Encapsulation?

Encapsulation – bundle data and the operations on that data together, and **hide internal implementation** from the outside world.

Two key ideas:

- **Data hiding** – only expose what consumers need
- **Controlled access** – use methods to validate changes

> It's the principle behind `private` fields and getters/setters.

Without Encapsulation

An interface is better than `any` – but it still can't enforce rules:

```
// TypeScript is happy; your data is not
const post: Post = { id: -99, title: "", content: "", author: "Alice" };

post.id = -99;    // ❌ Invalid ID — interface allows it
post.title = ""; // ❌ Empty title — interface allows it

// The object exists, but it's in a meaningless state
```

> Interfaces describe **what fields exist** – not **what values are valid**.

With Encapsulation

Using `public readonly` fields keeps the syntax concise while still enforcing rules in the constructor:

```
export class Post {
  constructor(
    public readonly id: number,
    public readonly title: string,
    public readonly author: string,
  ) {
    if (id <= 0) throw new Error("ID must be positive");
    if (!title.trim()) throw new Error("Title cannot be empty");
  }
}
```

`readonly` prevents reassignment after construction – fields can't be changed from outside.

> In larger codebases you may see `private` fields with getters – that gives finer control, but `public readonly` is simpler and idiomatic TypeScript.

Encapsulation & SOLID

Adding encapsulation keeps every other SOLID principle healthier:

SOLID Principle	How encapsulation helps
Single Responsibility	Validation lives in the model, not scattered across controllers
Open/Closed	Add new rules to the constructor without touching callers
Liskov Substitution	Subclasses can’t violate invariants hidden behind private fields
Dependency Inversion	Consumers depend on the interface, not the internal fields

Part 2

Models (MVC)

What is a Model?

In your MVC stack so far:

Layer	What it does
Route	Maps URL + HTTP method to a controller
Controller	Handles HTTP, validates input, calls service
Service	Fetches and persists data
Model	Defines the shape and rules of your data

The Model is the **single source of truth** for what your data looks like.

Two Types of Model

	Domain Model	DTO (Data Transfer Object)
Lives in	`models/`	`dtos/`
Purpose	Represents the real-world entity	Represents data sent over HTTP
Contains	Business rules, methods	Only raw data fields
Exposed to	Services only	Controllers and clients

> Your API response should **never** expose the domain model directly.

Why Separate Them?

Your internal `Post` model might have fields clients should never see:

```
// Domain Model — internal representation
export class Post {
  id: number;
  title: string;
  content: string;
  author: string;
  passwordHash: string;  // ❌ never send this to a client
  internalNotes: string; // ❌ internal use only
  createdAt: Date;
  updatedAt: Date;
}
```

Sending the full domain model leaks sensitive data and couples your API to your internals.

The DTO – What the Client Sees

```typescript
// src/dtos/postDto.ts
export interface PostResponseDto {
  id: number;
  title: string;
  author: string;
}

export interface CreatePostRequestDto {
  title: string;
  content: string;
  author: string;
}
```

> DTOs are plain data shapes – no methods, no private fields, no logic.

Domain Model as a Class

```ts
// src/models/post.ts
export class Post {
  constructor(
    public readonly id: number,
    public readonly title: string,
    public readonly content: string,
    public readonly author: string,
    public readonly createdAt: Date = new Date(),
  ) {
    if (!title.trim()) throw new Error("Title cannot be empty");
    if (!author.trim()) throw new Error("Author cannot be empty");
  }
}
```

The constructor **enforces invariants** – objects are always valid.

Models & SOLID

SOLID Principle	How models apply it
Single Responsibility	Domain model owns its own validation rules
Open/Closed	Add optional fields without breaking existing code
Liskov Substitution	A `Post` can always be used where a `Post` is expected
Interface Segregation	`PostResponseDto` only exposes fields the client needs
Dependency Inversion	Controller depends on `PostResponseDto`, not the class internals

Part 3

Putting it Together

Step 1 – Typed Service

Replace `any[]` in the service with your domain model class:

```typescript
// src/services/postService.ts
import { Post } from "../models/post";

const mockPosts: Post[] = [
  new Post(1, "Getting Started", "TypeScript basics...", "Alice"),
  new Post(2, "Advanced Patterns", "Generics explained...", "Bob"),
];

export class PostService {
  async findById(id: number): Promise<Post | undefined> {
    return mockPosts.find(p => p.id === id);
  }

  async findAll(): Promise<Post[]> {
    return mockPosts;
  }
}
```

The service now returns **guaranteed-valid** `Post` objects – invalid data can't be stored.

Step 2 – Controller Returns a DTO

The controller receives a `Post` from the service and builds a `PostResponseDto` to send:

```
import { Request, Response } from "express";
import { PostResponseDto } from "../dtos/postDto";
import { PostService } from "../services/postService";

export class PostController {
  constructor(private service: PostService = new PostService()) {}

  async getById(req: Request, res: Response): Promise<void> {
    const post = await this.service.findById(Number(req.params.id));
    if (!post) {
      res.status(404).json({ error: "Post not found" });
      return;
    }
    const dto: PostResponseDto = { id: post.id, title: post.title, author: post.author };
    res.status(200).json(dto);
  }
}
```

Only the fields declared in `PostResponseDto` are sent – sensitive fields are never exposed.

Step 3 – Creating from a Request DTO

Use `CreatePostRequestDto` to type incoming request bodies, then construct a `Post` :

```
async create(req: Request, res: Response): Promise<void> {
  const body: CreatePostRequestDto = req.body;

  // Post constructor validates — throws if title or author is empty
  const post = new Post(Date.now(), body.title, body.content, body.author); // temporary ID — real ID comes from the dat

  // In a real app, save to database here
  const dto: PostResponseDto = { id: post.id, title: post.title, author: post.author };
  res.status(201).json(dto);
}
```

Invalid data is **rejected at construction** – it never reaches the service or database.

Why This Follows SOLID

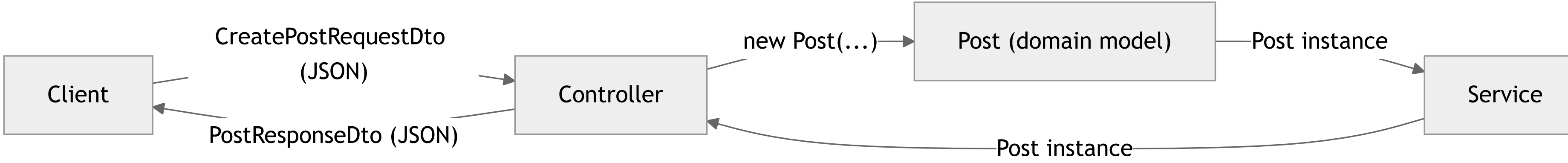
The create flow from the previous slide:

- **Single Responsibility** – `Post` validates, `PostService` stores, controller handles HTTP
- **Open/Closed** – add new fields to `Post` or `PostResponseDto` without touching other layers
- **Interface Segregation** – `CreatePostRequestDto` only carries fields the client should provide
- **Dependency Inversion** – controller depends on the DTO interface, not the internal `Post` class directly

Your Updated File Structure

```
src/
  index.ts                     ← entry point
  controllers/
    postController.ts          ← HTTP handling, builds DTOs, calls service
  routes/
    postRouter.ts              ← thin routes
  services/
    postService.ts             ← data access, works with Post instances
  models/
    post.ts                    ← domain model class with validation
  dtos/
    postDto.ts                 ← PostResponseDto, CreatePostRequestDto
dist/                          ← compiled output (git-ignored)
```

How the Layers Flow



> The domain model **never leaves the backend** – the controller extracts only what the DTO declares.

Summary

- `any` – disables type checking; avoid it as soon as you know the shape of your data
- **Interfaces** – give data a known shape; use them for DTOs and contracts
- **Classes** – add validation and encapsulation; use them for domain models
- **Domain Models** – represent real entities with rules enforced in the constructor
- **DTOs** – control exactly what enters and leaves your API
- Together: invalid data is rejected early, sensitive fields never leak, every layer has one clear job

> **Scaling up?** In larger projects with many DTOs, extract the mapping into a helper function (e.g. `toPostResponseDto(post)`) to keep controllers clean.

Now it's your turn

Exercise 4 – Refactor your expense API:

1. Convert `Expense` from `any` to a typed interface, then to a class with constructor validation
2. Create `ExpenseResponseDto` and `CreateExpenseRequestDto`
3. Update `ExpenseService` to store and return `Expense` class instances
4. Update `ExpenseController` to build an `ExpenseResponseDto` before sending the response